Invited Paper

Optimization of cloak effect in nanosphere dispersed liquid crystal metamaterial

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ABSTRACT

In this paper the concept and design of infrared cloaking using nanosphere dispersed liquid crystal (NDLC) matematerial in cylindrical geometry is presented for TM polarization of incident light. The influence of material losses on the cloaking efficiency is discussed. The loss can be tuned by changing design parameters.

1. CLOAKING EFFECT – AN OUTLINE

In 1968 by Veselago in his seminal paper[1] has introduced the concept of metamaterials by showing that Maxwell equations allow for the existence of propagating solutions corresponding to negative values of refractive index. However, this concept was not further explored because materials with such unusual properties can not be found in nature. The interest in the field of metamaterials started to grow steadily since the first experimental realization by Pendry[2] in 1998. The used metamaterial, composed of thin metal wires and split-ring resonators, offered no possibility of tuning. Since the first experimental realization of metamaterial behavior many different approaches to obtain the negative effective refractive index were invented. An incomplete survey of some of the recent result can be found in a review paper.[3]

In order to introduce the tunability in the metamaterial system one can make use of the birefringent properties of liquid crystals. Nanosphere dispersed liquid crystal (NDLC) metamaterial[4] offers a possibility of tuning of refractive index by changing the spatial distribution of the orientations of nematic liquid crystal (NLC) molecules which, in turn, causes the change of the permittivity of the host and influences the effective parameters of the metamaterial. We have applied Monte Carlo simulations to study the perspectives for tunability resulting from a joint action of electrostatic field, anchoring forces and temperature.[5–7]

Recently we have proposed a design of an invisibility cloak using NDLC metamaterial.[8–10] The aim of this paper is to discuss the influence of metamaterial losses ($Im\{\varepsilon_{eff}\} = \varepsilon''_{eff}$) on the cloaking efficiency of the layer-like design of invisibility cloak proposed recently in Ref.[10]

2. NDLC METAMATERIAL

In this Section we review briefly the concept of NDLC metamaterial following the presentation from Ref.[3]

The studied NDLC metamaterial[4,11] is based on nematic liquid crystal in which the coated nonmagnetic nanospheres with core-shell structure are randomly distributed (Fig. 1). In this material the effective refractive index depends on the composition and dimensions of the coated spheres (Fig. 1), filling factor, and on the angle $\theta(\vec{r})$ which the long axis of the NLC molecule at point $\vec{r}$ makes with the direction of probing light. The core with radius r_1 is made from polaritonic material with relative permittivity

$$\varepsilon_1(\omega) = \varepsilon_1(\infty)\left(1 + \frac{\omega_L^2 - \omega_T^2}{\omega_T^2 - \omega^2 - i\omega\gamma_1}\right), \tag{1}$$

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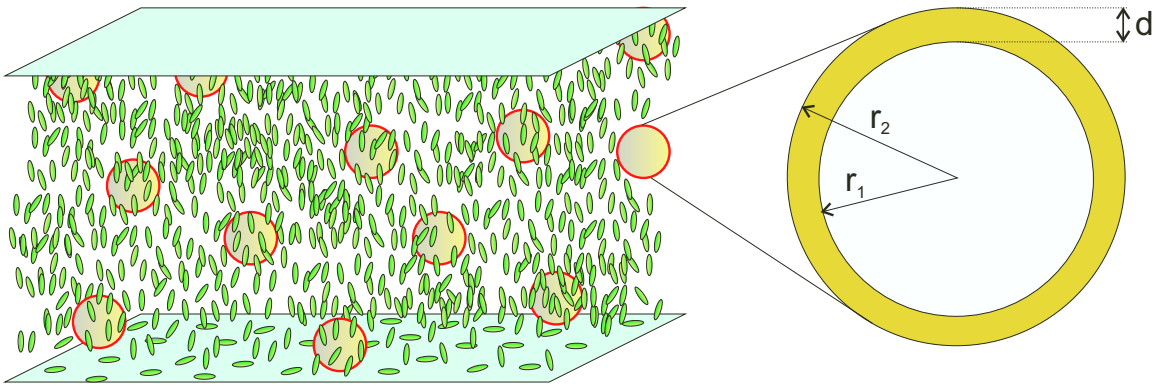

Figure 1. Model of the system: liquid crystal and dispersed nanospheres (left). Coated nanosphere (right).

where $\varepsilon_1(\infty)$ denotes the high-frequency dielectric constant, γ_1 – damping coefficient, ω_T and ω_L – transverse and longitudinal phonon frequencies, respectively. The relative permittivity of the semiconductor shell with outer radius $r_2 = r_1 + d$ is described by Drude model:

$$\varepsilon_2(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\omega\gamma_2}, \tag{2}$$

where ω_p denotes plasma frequency and γ_2 – the damping coefficient.

Permittivity of the liquid crystal host for extraordinary polarized wave depends on angle θ:[11]

$$\varepsilon_3(\theta) = \frac{\varepsilon_\parallel \varepsilon_\perp}{\varepsilon_\parallel \cos^2(\theta) + \varepsilon_\perp \sin^2(\theta)}, \tag{3}$$

where $\varepsilon_\parallel$ and $\varepsilon_\perp$ are the relative permittivities for light polarized respectively parallel and perpendicular to the director axis.

The effective permittivity ε_{eff} and permeability μ_{eff} of NDLC system were calculated using the method presented in papers[4,12] in the formalism of Mie scattering and Maxwell Garnet mixing rule:[13–15]

$$\varepsilon_{eff} = \varepsilon_3 \left(\frac{k_3^3 + 4\pi i N a_1}{k_3^3 - 2\pi i N a_1} \right), \ \mu_{eff} = \frac{k_3^3 + 4\pi i N b_1}{k_3^3 - 2\pi i N b_1}, \tag{4}$$

where a_1 and b_1 are scattering coefficients,[13] $k_3 = \sqrt{\varepsilon_3}\, 2\pi/\lambda$, λ denotes the free-space wavelength. N denotes number density of the spheres: $N = m/V_u$, where m stands for the number of spheres in a unit volume V_u. Formula (4) is valid in long-wavelength limit $\lambda/(\mathrm{r}_1+\mathrm{d})>>1$, for low values of filling factor f.[4]

3. INFRARED CLOAK IN NDLC

3.1 Cloak design in the absence of losses

In a recent paper[8] we have designed an infrared cylindrical TM cloak based on NDLC in the absence of losses. Consider a cylindrical NLC cell with inner and outer radii a and b, filled with NDLC metamaterial, with planar azimuthal alignment of NLC molecules in $x - y$ plane. The incident light with transverse magnetic (TM) polarization ($\vec{E}$ in $x - y$ plane and $\vec{B}$ along the axis of the cylinder) propagates in $x - y$ plane, see Ref.[8] for details.

The research done in Ref.[16] leads to the conclusion that for the cloak design working only for TM polarization it suffices to ensure proper values of the products $(\mu'_{eff})_z(\varepsilon'_{eff})_\Theta$ and $(\mu'_{eff})_z(\varepsilon'_{eff})_r$ (here Θ and r denote polar coordinates in $x - y$ plane, and primes - real parts):

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$$(\mu'_{eff})_z(\varepsilon'_{eff})_r = \left(\frac{b}{b-a}\right)^2\left(\frac{r-a}{r}\right)^2, \tag{5}$$

$$(\mu'_{eff})_z(\varepsilon'_{eff})_\Theta = \left(\frac{b}{b-a}\right)^2. \tag{6}$$

The tensor components of $\varepsilon'_{eff}, \mu'_{eff}$ are calculated using the corresponding tensor components of ε_3. In cylindrical coordinates the tensor ε_3 corresponding to planar azimuthal alignment of NLC molecules in $x-y$ plane is diagonal with components $\varepsilon_r = \varepsilon_z = \varepsilon_\perp = n_o^2, \varepsilon_\Theta = \varepsilon_\parallel = n_e^2$. Real parts of radial and azimuthal components of effective parameters were calculated from Eq. (4): $(\varepsilon'_{eff})_r$ was calculated using $\varepsilon_3 = \varepsilon_r$ and $(\varepsilon'_{eff})_\Theta$ – using $\varepsilon_3 = \varepsilon_\Theta$.

The ultimate goal of the design was to find the variation in space of NLC host parameters n_o, n_e inside the cloak domain which guarantee that cloaking relations (5) – (6) hold, see Ref.[8] for details.

3.2 Lossess: perspectives

An important factor in a practical cloak design is loss associated with the imaginary part of the complex refractive index.[20] We try to examine absorption properties of the cloak, which allow to obtain losses reduction in comparison with shadow behind the object without the cloak. For chosen size of the hidden object and the cloak we assume homogeneous distribution of imaginary part of the permittivity. Real parts of electromagnetic tensor components are taken from eqs (5, 6). To qualitatively evaluate absorption properties of the cloak device we have made field-mapping calculations using Finite Element package (COMSOL Multiphysics). In Fig. 2 the Poynting vector distribution behind the object is presented. In those calculations losses related to absorption in the cloak were not taken into account.

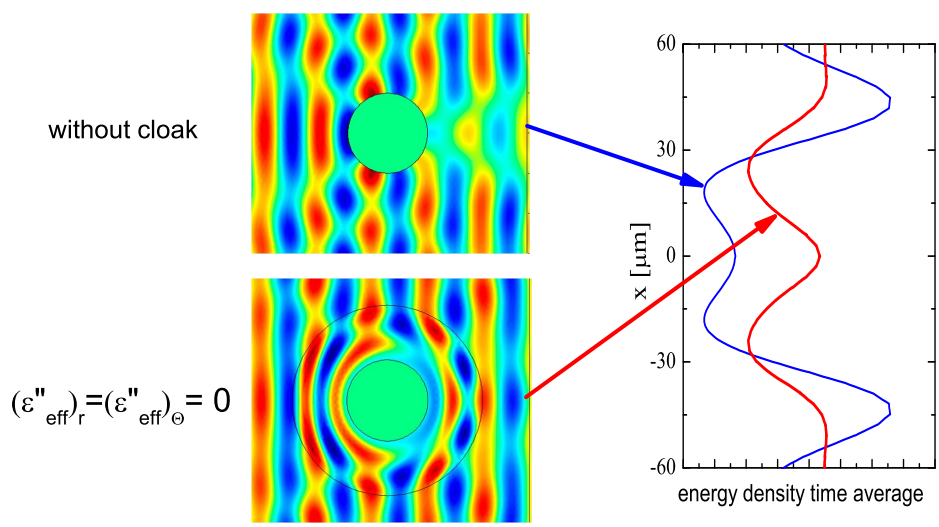

Figure 2. Field-mapping calculations using Finite Element package (COMSOL Multiphysics). The object hidden inside the cloak is an ideal metallic cylinder with radius $a = 20\,\mu$m. The outgoing waves were absorbed at all the boundaries in the simulation domain. Without the NDLC shell a shadow is cast behind the cylinder (top) while in the presence of the cloak (bottom) the wave fronts flow around the cloaked region with a rather small perturbation. $\lambda = 27.7\,\mu$m, $b = 46.9\,\mu$m.

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On the contrary in Fig. 3 the Poynting vector distribution with losses related to imaginary part of the permittivity is presented for $(\varepsilon''_{eff})_\Theta = (\varepsilon''_{eff})_r = 0.1$ and 0.01. Analysis of Poynting vector distribution shows that the value 0.1 is prohibitively large for the used sizes and wavelength.

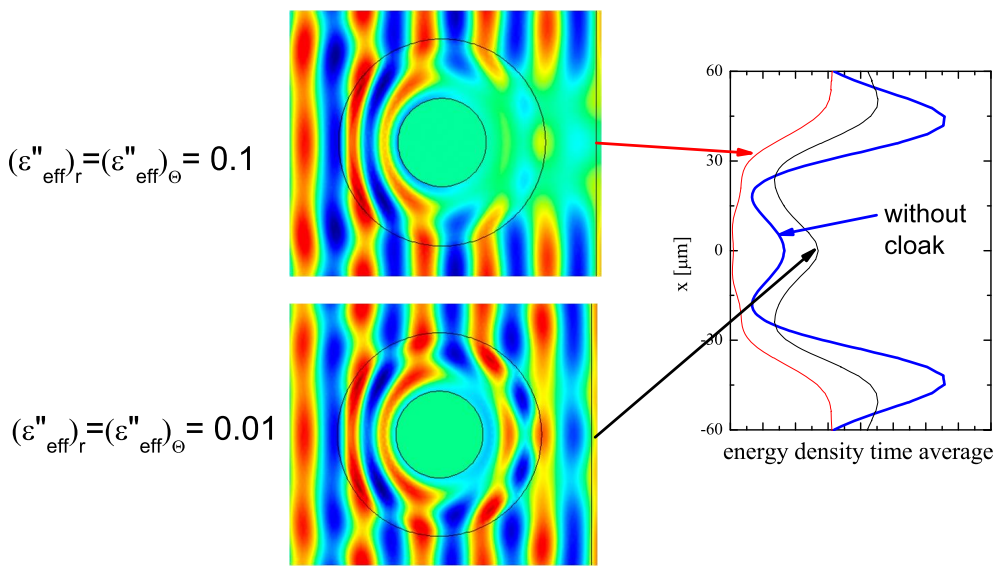

Figure 3. Field-mapping calculations using Finite Element package (COMSOL Multiphysics) for $(\varepsilon''_{eff})_\Theta = (\varepsilon''_{eff})_r = 0.1$ (top) and 0.01 (bottom). $\lambda = 27.7\,\mu$m, $b = 46.9\,\mu$m.

Figure 4 shows the plots of an imaginary part of the effective permittivity of NDLC metamaterial calculated using parameters from the original paper[4] (curve 1), in the second case (curve 2), with a changed plasma frequency of 15 THz and $r_2 = 800\,\mu$m, $\lambda = 27.7\,\mu$m and for $f = 0.12$, $r_2 = 500\,\mu$m, $\lambda = 25\,\mu$m (curve 3). To avoid prohibitively large losses the imaginary part ε'' has to be sufficiently small: $\varepsilon'' < 0.1$. The loss optimization, which requires systematic massive computation efforts, is at progress now; the results will be presented in forthcoming publications.

4. CONCLUSIONS

We have shown that the efficiency of the cloak with low value of the imaginary part of permittivity ($\varepsilon''_{eff} = 0.01$) is close to the lossless case. On the other hand the case with ($\varepsilon''_{eff} = 0.1$) results in the shadow even greater than the object itself (without the cloak). The loss can be minimized by scanning the multidimensional space of design parameters.

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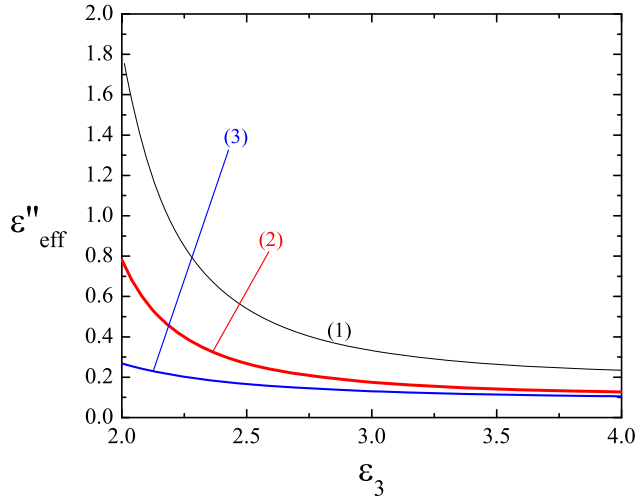

Figure 4. Imaginary part of effective permittivity as a function of ε_3: calculated using the parameters from Ref.[4] (curve 1), calculated using changed plasma frequency 15 THz and $r_2 = 800\,\mu$m (curve 2) and for $f = 0.12$, $r_2 = 500\,\mu$m, $\lambda = 25\,\mu$m.

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